Set A

Daffodil International University

Dept. of SWE

Quiz 1

Course Code: SE 321

Course Title: Software Engineering Web Application

Time: 45 minutes **Full Marks:** 15

Name: _____ **Student ID:** _____

Instructions: Answer all questions. Show your steps for the HTTP timing calculation. Write HTML and CSS clearly in the space provided.

Question 1: Persistent and Non-Persistent HTTP (5 marks)

A web page consists of 1 HTML file and 3 image files (4 objects in total). Assume the browser can request all four objects, the round-trip time (RTT) is 100 ms, and transmitting each object takes 20 ms. Count one RTT for a TCP connection setup and one RTT for each HTTP request/response. Ignore any other delay.

(a) Calculate the total time to fetch all objects using non-persistent HTTP. Show the formula and calculation. (2 marks)

(b) Calculate the total time using persistent HTTP over one reused TCP connection. Show the formula and calculation. (2 marks)

(c) How much time is saved by persistent HTTP in this example? (1 mark)

Question 2: HTTP, URL and DNS (3 marks)

Write two differences among HTTP, a URL and DNS. Then give one example showing how they work together when someone opens a website. (3 marks)

Question 3: HTML and CSS Form (5 marks)

Write HTML and CSS code for a simple student registration form. Include a heading, labeled fields for the student name and email, a program/department selection list, and a submit button. Use suitable input types and make the form neat with basic CSS.

Suggested marking: HTML structure and fields (3), labels and suitable controls (1), CSS styling (1).

Question 4: Web Request and Response Scenario (2 marks)

A student submits a registration form in a web application. The browser sends the request to the application server/API, which processes it and returns a response. Draw a diagram showing this interaction. Label the user/browser, server/API, and communication protocol. Briefly explain how the request and response move. (Diagram: 1.5 marks; explanation: 0.5 mark)